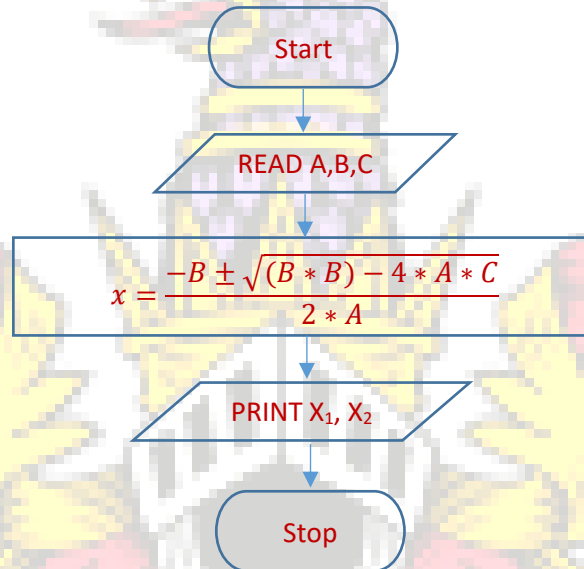




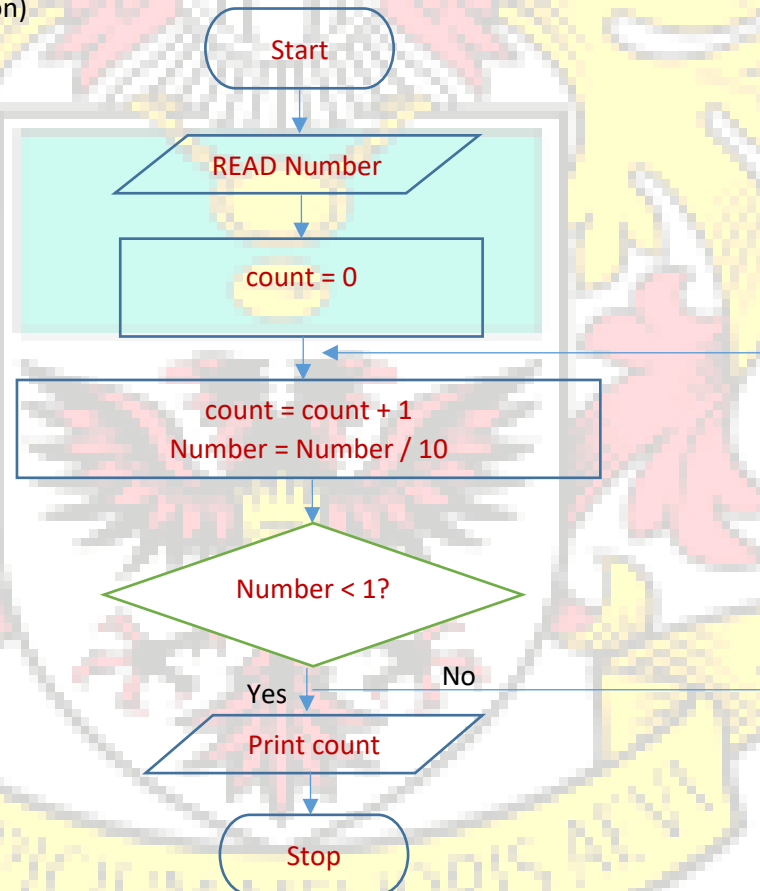
Tutorial S1A: Algorithmic Representation (Solutions)

Section A (Discussion Questions)

1. (Sequence)

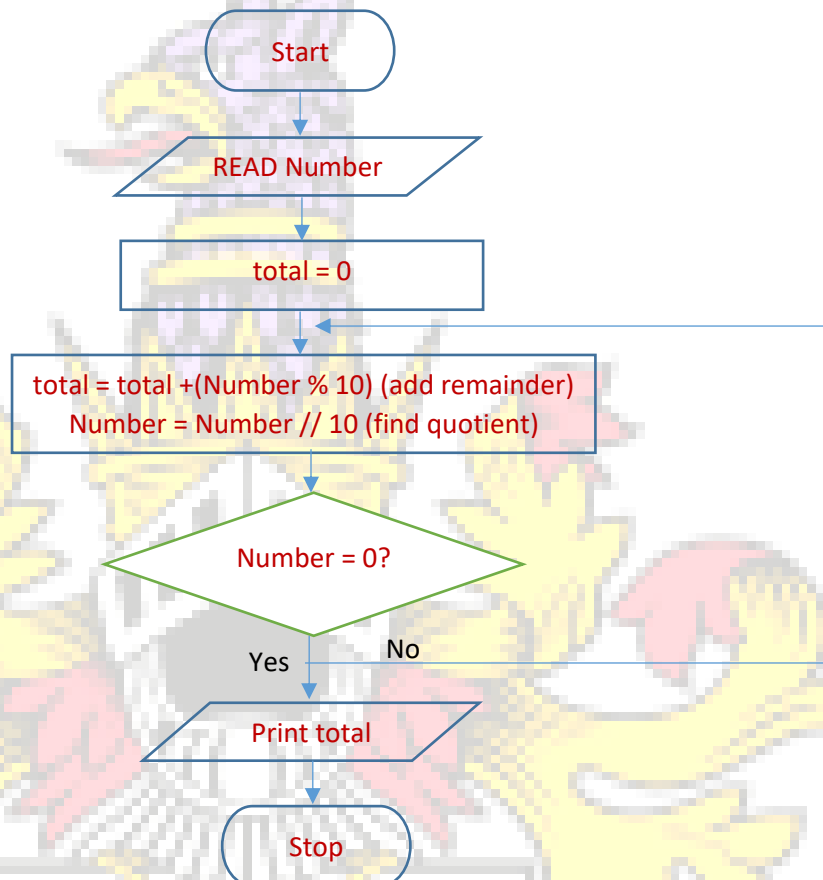


2. (iteration/repetition)

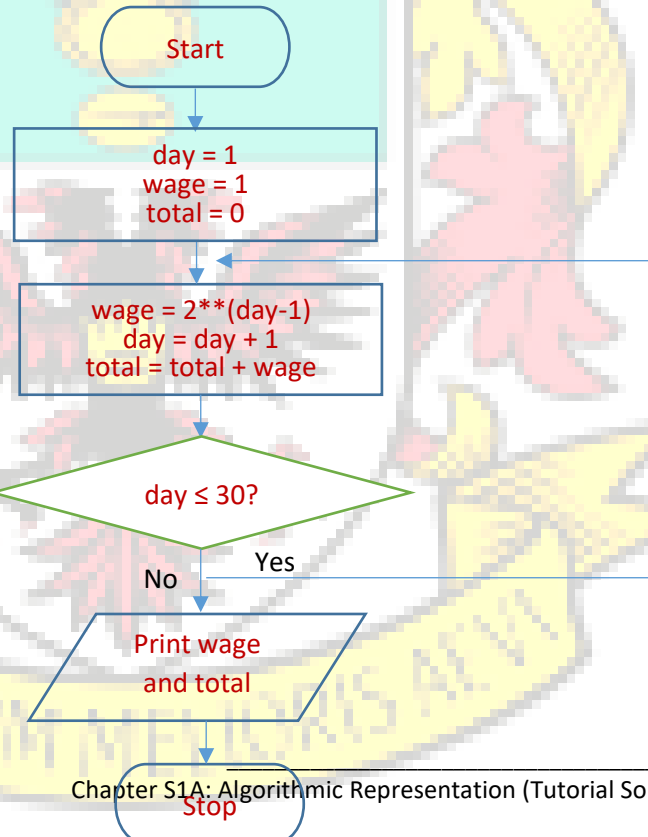




3. (iteration/repetition)



4. (iteration – count-controlled)



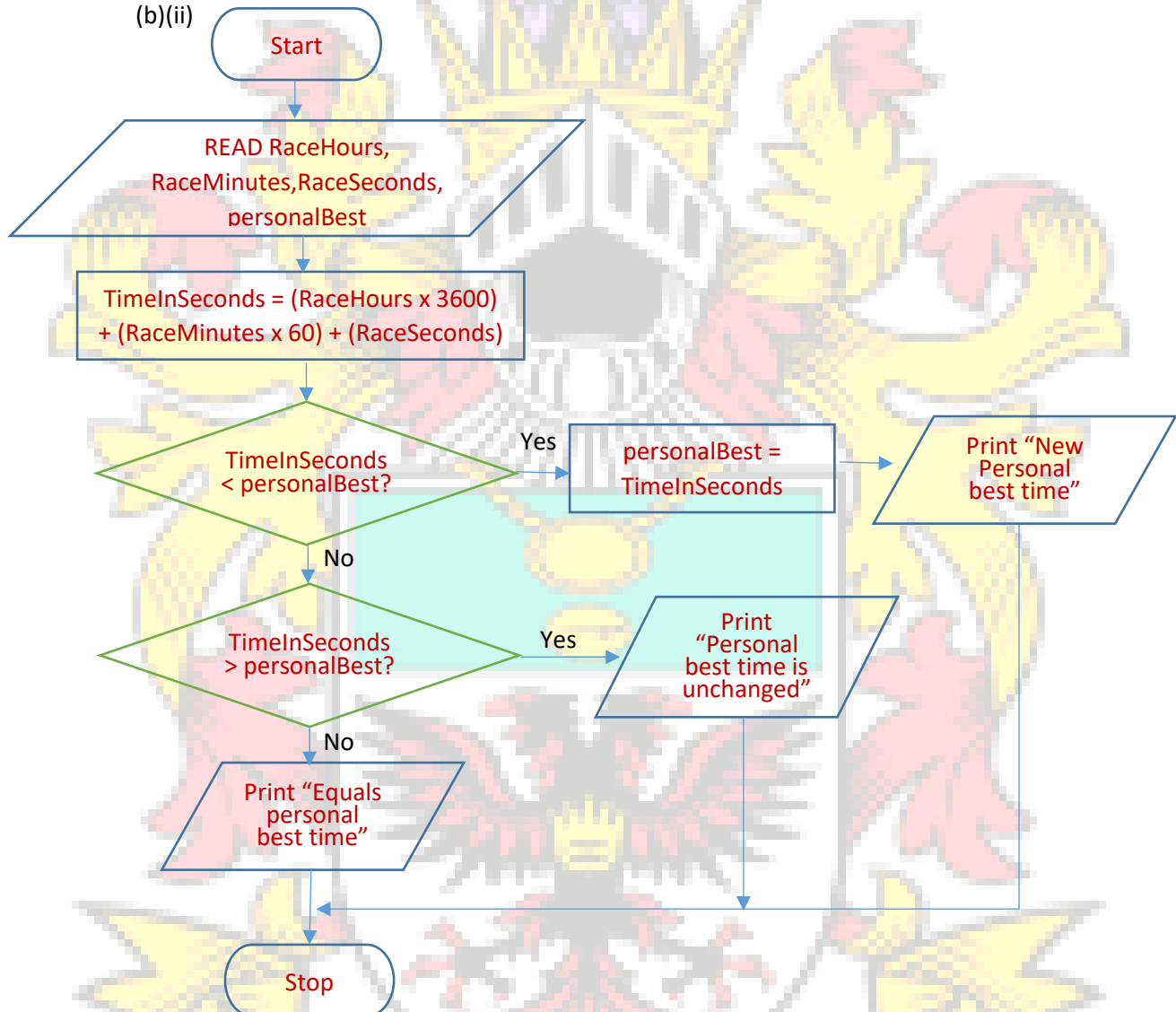


5. (a)

Identifier	Description
RaceHours	The hours part of the race time.
RaceMinutes	Minutes part of race time
RaceSeconds	Seconds part of race time
TimeInSeconds	Race time converted to seconds

(b)(i) personalBest : Personal best time of runner

(b)(ii)





6. (a)

Data type	Identifier	Description
STRING	employeeID	Consists of several letters and numbers
REAL	salary	Consists of number with fractional part
CHAR	gender	Only a single letter to represent, 'M' or 'F'
INTEGER		Accept any suitable answer
BOOLEAN		Accept any suitable answer

(b)(i) Procedures: do not return a value after running the statements in the subroutine/subprogram.
Functions: return a value after running

(b)(ii) Calculate total tax payable will return the **value of tax payable** to the calling program, so that the tax payable may be used for other modules of the program, such as to be used to subtract from the gross pay. Therefore use a function.
Output list of employees in alphabetical order will not need to return any value to the calling program. Therefore use a procedure.

(c) pass-by-value: passes a copy of the actual value of the variable, original value is not changed by the subroutine/ subprogram

pass-by-reference: passes the address of the variable, original value may be changed in the subroutine/ subprogram

7.

CONDITIONS	OUTCOMES			
	1	2	3	4
Purchase price of goods $\geq$ \$2000?	Y	Y	N	N
Delivery distance < 20km?	Y	N	Y	N
ACTIONS				
Customer gets free delivery	X			
Charge customer 5% of price		X	X	
Charge customer 10% of price				X

8.

(a)

Conditions	1	2	3	4	5	6	7	8
In loyalty prog 5-9 yrs?	Y	Y	Y	Y	N	N	N	N
In loyalty prog >10 yrs?	Y	Y	N	N	Y	Y	N	N
Cumulative value for calendar yr >\$1000?	Y	N	Y	N	Y	N	Y	N
Actions								
Receive 5% discount	*	*		X				
Receive 10% discount	*	*				X		
Receive 15% discount	*	*	X		X		X	

(b)

Conditions	1	2	3
In loyalty prog 5-9 yrs?	-	Y	N
In loyalty prog >10 yrs?	-	N	Y
Cumulative value for calendar yr >\$1000?	Y	N	N
Actions			
Receive 5% discount		X	
Receive 10% discount			X
Receive 15% discount	X		



9 (a)	OUTCOMES							
CONDITIONS	1	2	3	4	5	6	7	8
Withdrawal amount $\leq$ balance?	Y	Y	Y	Y	N	N	N	N
Withdrawal amount $\leq$ daily limit?	Y	Y	N	N	Y	Y	N	N
Current balance $<$ \$50?	Y	N	Y	N	Y	N	Y	N
ACTIONS								
Successful transaction	X	X						
Reject transaction			X	X	X	X	X	X
Fix charge incurred	X							

(b)	OUTCOMES			
CONDITIONS	1	2	3	4
Withdrawal amount $\leq$ balance?	Y	Y	-	N
Withdrawal amount $\leq$ daily limit?	Y	Y	N	-
Current balance $<$ \$50?	Y	N	-	-
ACTIONS				
Successful transaction	X	X		
Reject transaction			X	X
Fix charge incurred	X			

```

(c)  FUNCTION WITHDRAW(amount: INTEGER) RETURNS INTEGER
      IF amount  $\leq$  balance AND amount  $\leq$  limit
      THEN
        IF balance  $<$  50
        THEN
          RETURN charge // charge for a chargeable successful transaction
        ELSE
          RETURN 0 // charge-free successful transaction
        ENDIF
      ELSE
        RETURN -1 // rejected transaction
      ENDIF
    ENDFUNCTION
  
```

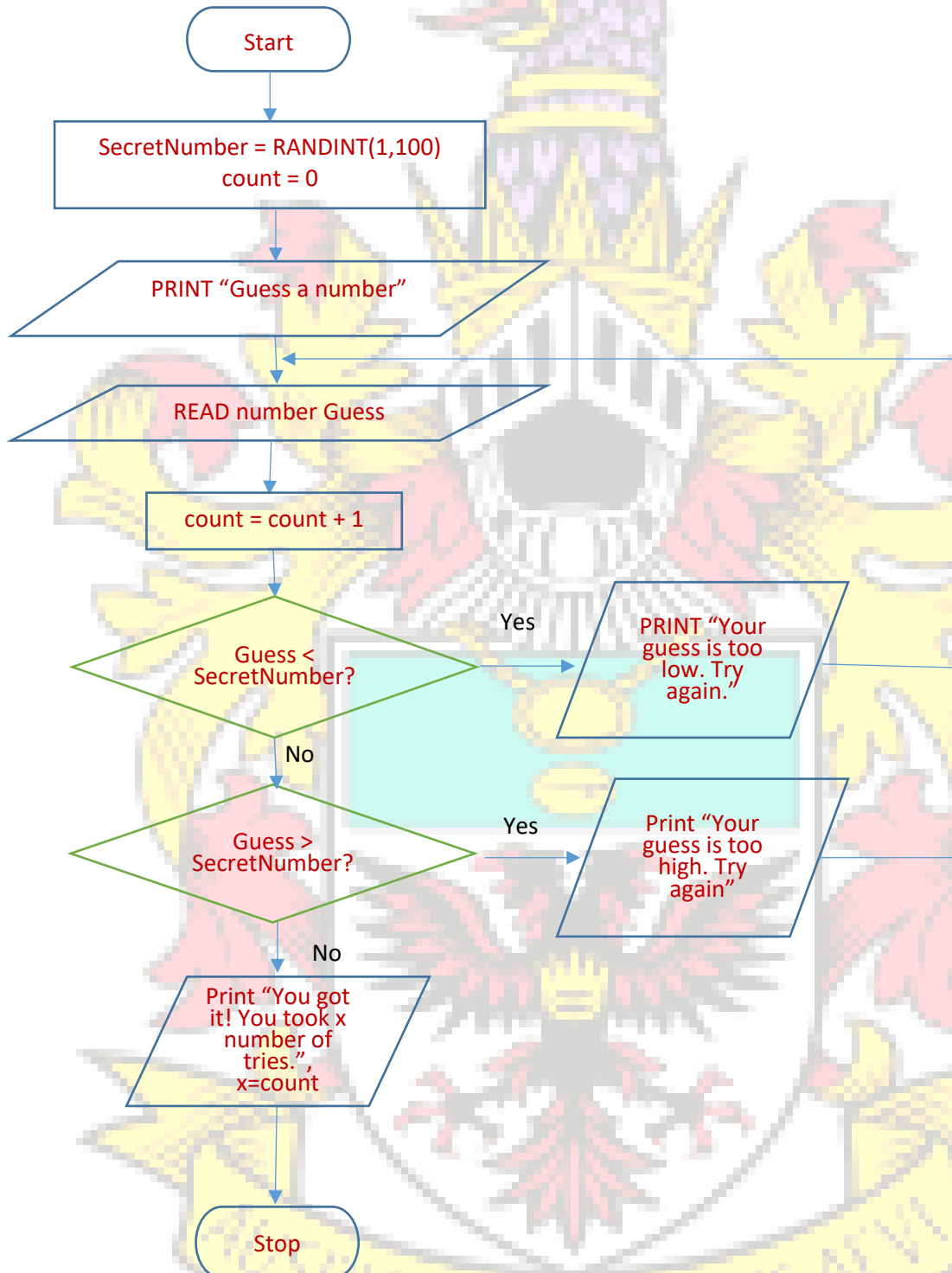
10 (a)	Condition	1	2	3	4	5	6	7	8
	Book overdue by more than 5 days?	Y	Y	Y	Y	N	N	N	N
	Student has had a previous warning letter?	Y	Y	N	N	Y	Y	N	N
	More than 4 books overdue?	Y	N	Y	N	Y	N	Y	N
Action									
Send student reminder letter					X	X	X	X	X
Send student warning letter		X	X	X					
Send student's parent copy of warning letter		X	X						

(b)	Condition	1	2	3	4
	Book overdue by more than 5 days?	Y	Y	Y	N
	Student has had a previous warning letter?	Y	N	N	-
	More than 4 books overdue?	-	Y	N	-
Action					
Send student reminder letter				X	X
Send student warning letter		X	X		
Send student's parent copy of warning letter		X			



Section B (Assignment Questions)

1. (flowchart – number guessing game)





2. (a) (i) "Number not valid"
 (a) (ii) 1, 3, 6, 10, 15
 (b) Amend this statement: "Total $\leftarrow$ Total + i" to "Total $\leftarrow$ Total + i x i"
 3. (a)

Condition	OUTCOMES							
	1	2	3	4	5	6	7	8
Has customer been refused insurance by another company?	Y	Y	Y	Y	N	N	N	N
Is car over 10 years old?	Y	Y	N	N	Y	Y	N	N
Has customer made more than 3 claims previously?	Y	N	Y	N	Y	N	Y	N
Action								
Insurance refused	X	X						
Insurance without discount available			X	X	X	X		
Insurance with discount available								X

* denotes rule is not possible because conditions not explicitly stated in question

(b)

Condition	OUTCOMES			
	1	2	3	4
Has customer been refused insurance by another company?	Y	Y	N	N
Is car over 10 years old?	Y	N	Y	N
Has customer made more than 3 claims previously?	-	-	-	N
Action				
Insurance refused	X			
Insurance without discount available		X	X	
Insurance with discount available				X